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<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Advanced System Dynamics Dashboard</title>

<script src="https://cdn.plot.ly/plotly-latest.min.js"></script>

<style>

:root{
  --sidebar-width:300px;
  --plot-height:150px;
  --card-radius:8px;
  --border:#d9d9d9;
  --bg:#f7f7f7;
}

*{
  box-sizing:border-box;
}

body{
  margin:0;
  font-family:Arial,sans-serif;
  background:white;
  overflow:hidden;
}

/* HEADER */

header{
  height:56px;
  display:flex;
  align-items:center;
  justify-content:space-between;
  padding:0 12px;
  border-bottom:1px solid var(--border);
  background:white;
}

header h1{
  margin:0;
  font-size:18px;
  font-weight:700;
  color:#111;
}

.toolbar{

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        display:flex;
        gap:6px;
        align-items:center;
    }

    .btn{
        border:1px solid var(--border);
        background:white;
        padding:6px 10px;
        border-radius:6px;
        cursor:pointer;
        font-size:12px;
        font-weight:bold;
    }

    .btn:hover{
        background:#f0f0f0;
    }

    /* MAIN LAYOUT */

    .main-layout{
        display:flex;
        height:calc(100vh - 56px);
    }

    /* SIDEBAR */

    .sidebar{
        width:var(--sidebar-width);
        background:var(--bg);
        border-right:1px solid var(--border);
        overflow-y:auto;
        padding:8px;
    }

    .sidebar-section{
        margin-bottom:8px;
        background:white;
        border:1px solid var(--border);
        border-radius:var(--card-radius);
    }

    .sidebar-section summary{
        padding:8px;
        cursor:pointer;
        font-size:12px;
        font-weight:bold;
        background:#fafafa;
    }

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.slider-box{
  padding:6px 8px;
  border-top:1px solid #f0f0f0;
}

.slider-box label{
  display:block;
  font-size:10px;
  font-weight:bold;
  margin-bottom:2px;
  color:#333;
}

.slider-box input{
  width:100%;
}

.slider-value{
  font-size:10px;
  color:#555;
  margin-top:2px;
}

/* MAIN CONTENT */

.content{
  flex:1;
  overflow-y:auto;
  padding:8px;
}

/* KPI BAR */

.kpi-row{
  display:grid;
  grid-template-columns:repeat(6,1fr);
  gap:6px;
  margin-bottom:8px;
}

.kpi-card{
  background:white;
  border:1px solid var(--border);
  border-radius:var(--card-radius);
  padding:8px;
}

.kpi-title{
  font-size:10px;
  color:#666;
  margin-bottom:4px;
}

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}

.kpi-value{
    font-size:18px;
    font-weight:bold;
    color:#111;
}

/* TABS */

.tabs{
    display:flex;
    gap:4px;
    margin-bottom:8px;
    flex-wrap:wrap;
}

.tab-btn{
    border:1px solid var(--border);
    background:white;
    padding:6px 10px;
    border-radius:6px;
    cursor:pointer;
    font-size:11px;
    font-weight:bold;
}

.tab-btn.active{
    background:#111;
    color:white;
}

/* PLOT GRID */

.plot-grid{
    display:grid;
    grid-template-columns:repeat(4,1fr);
    gap:6px;
}

.plot-card{
    background:white;
    border:1px solid var(--border);
    border-radius:var(--card-radius);
    overflow:hidden;
}

.plot-title{
    font-size:10px;
    font-weight:bold;
    padding:5px 8px;
}

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    border-bottom:1px solid #f0f0f0;
    background:#fafafa;
}

.plot{
    height:var(--plot-height);
}

/* DOCUMENTATION */

.docs{
    margin-top:8px;
    border:1px solid var(--border);
    border-radius:var(--card-radius);
    background:white;
}

.docs summary{
    padding:8px;
    font-size:12px;
    font-weight:bold;
    cursor:pointer;
}

.docs-content{
    padding:10px;
    font-size:12px;
    line-height:1.5;
    border-top:1px solid #f0f0f0;
}

/* RESPONSIVE */

@media(max-width:1600px){

    .plot-grid{
        grid-template-columns:repeat(3,1fr);
    }

}

@media(max-width:1200px){

    .plot-grid{
        grid-template-columns:repeat(2,1fr);
    }

    .kpi-row{
        grid-template-columns:repeat(3,1fr);
    }

}

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}

@media(max-width:900px){

    .main-layout{
        flex-direction:column;
    }

    .sidebar{
        width:100%;
        height:300px;
    }

    .plot-grid{
        grid-template-columns:1fr;
    }

}

</style>
</head>

<body>

<header>

<h1>Advanced System Dynamics Dashboard</h1>

<div class="toolbar">

<button class="btn" onclick="runSimulation()">▶ Run</button>
<button class="btn" onclick="pauseSimulation()">⏸ Pause</button>
<button class="btn" onclick="resetSimulation()">↺ Reset</button>
<button class="btn" onclick="randomizeParameters()">⚡ Randomize</button>
<button class="btn" onclick="exportData()">📄 Export</button>

<div style="font-size:12px;font-weight:bold;padding-left:8px;">
Day: <span id="simDay">0</span>
</div>

</div>

</header>

<div class="main-layout">

<!-- SIDEBAR -->

<div class="sidebar">

<details class="sidebar-section" open>

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<summary>Healthcare Parameters</summary>

<div class="slider-box">
<label>Elective Capacity</label>
<input type="range" min="0" max="500" value="267" id="s1">
<div class="slider-value" id="s1v">267</div>
</div>

<div class="slider-box">
<label>Bed Capacity</label>
<input type="range" min="0" max="8000" value="5600" id="s2">
<div class="slider-value" id="s2v">5600</div>
</div>

<div class="slider-box">
<label>Emergency Admissions</label>
<input type="range" min="0" max="150" value="60" id="s3">
<div class="slider-value" id="s3v">60</div>
</div>

<div class="slider-box">
<label>Length of Stay</label>
<input type="range" min="1" max="40" value="23" id="s4">
<div class="slider-value" id="s4v">23</div>
</div>

</details>

<details class="sidebar-section" open>
<summary>Workforce Parameters</summary>

<div class="slider-box">
<label>Burnout Sensitivity</label>
<input type="range" min="0" max="1" step="0.01" value="0.35" id="s5">
<div class="slider-value" id="s5v">0.35</div>
</div>

<div class="slider-box">
<label>Productivity Loss</label>
<input type="range" min="0" max="0.5" step="0.01" value="0.20" id="s6">
<div class="slider-value" id="s6v">0.20</div>
</div>

<div class="slider-box">
<label>Staff Retention</label>
<input type="range" min="0" max="1" step="0.01" value="0.75" id="s7">
<div class="slider-value" id="s7v">0.75</div>
</div>

</details>

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<details class="sidebar-section">
<summary>Population Dynamics</summary>

<div class="slider-box">
<label>Population Growth</label>
<input type="range" min="0" max="5" step="0.1" value="1.2" id="s8">
<div class="slider-value" id="s8v">1.2</div>
</div>

<div class="slider-box">
<label>Ageing Pressure</label>
<input type="range" min="0" max="1" step="0.01" value="0.55" id="s9">
<div class="slider-value" id="s9v">0.55</div>
</div>

</details>

<details class="sidebar-section">
<summary>Economic Parameters</summary>

<div class="slider-box">
<label>Inflation</label>
<input type="range" min="0" max="15" step="0.1" value="3" id="s10">
<div class="slider-value" id="s10v">3</div>
</div>

<div class="slider-box">
<label>Government Funding</label>
<input type="range" min="0" max="100" value="50" id="s11">
<div class="slider-value" id="s11v">50</div>
</div>

</details>

<details class="sidebar-section">
<summary>Climate & Energy</summary>

<div class="slider-box">
<label>Energy Shock</label>
<input type="range" min="0" max="1" step="0.01" value="0.2" id="s12">
<div class="slider-value" id="s12v">0.2</div>
</div>

<div class="slider-box">
<label>Climate Stress</label>
<input type="range" min="0" max="1" step="0.01" value="0.3" id="s13">
<div class="slider-value" id="s13v">0.3</div>
</div>

</details>

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</div>

<!-- MAIN CONTENT -->

<div class="content">

  <!-- KPI ROW -->

  <div class="kpi-row">

    <div class="kpi-card">
      <div class="kpi-title">Occupied Beds</div>
      <div class="kpi-value" id="kpi1">5,230</div>
    </div>

    <div class="kpi-card">
      <div class="kpi-title">Waiting List</div>
      <div class="kpi-value" id="kpi2">212k</div>
    </div>

    <div class="kpi-card">
      <div class="kpi-title">Burnout Index</div>
      <div class="kpi-value" id="kpi3">0.42</div>
    </div>

    <div class="kpi-card">
      <div class="kpi-title">Occupancy</div>
      <div class="kpi-value" id="kpi4">93%</div>
    </div>

    <div class="kpi-card">
      <div class="kpi-title">Admissions</div>
      <div class="kpi-value" id="kpi5">78</div>
    </div>

    <div class="kpi-card">
      <div class="kpi-title">Delayed Discharge</div>
      <div class="kpi-value" id="kpi6">840</div>
    </div>

  </div>

  <!-- TABS -->

  <div class="tabs">
    <button class="tab-btn active">Core Dynamics</button>
    <button class="tab-btn">Healthcare</button>
    <button class="tab-btn">Economy</button>
    <button class="tab-btn">Climate</button>
    <button class="tab-btn">Population</button>
    <button class="tab-btn">Policy</button>
  </div>

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<button class="tab-btn">Sensitivity</button>
<button class="tab-btn">Scenarios</button>
</div>

<!-- PLOTS -->

<div class="plot-grid">

  <div class="plot-card">
    <div class="plot-title">Waiting List</div>
    <div id="plot1" class="plot"></div>
  </div>

  <div class="plot-card">
    <div class="plot-title">Occupied Beds</div>
    <div id="plot2" class="plot"></div>
  </div>

  <div class="plot-card">
    <div class="plot-title">Burnout</div>
    <div id="plot3" class="plot"></div>
  </div>

  <div class="plot-card">
    <div class="plot-title">Admissions</div>
    <div id="plot4" class="plot"></div>
  </div>

  <div class="plot-card">
    <div class="plot-title">Discharge Delay</div>
    <div id="plot5" class="plot"></div>
  </div>

  <div class="plot-card">
    <div class="plot-title">Acute Inpatients</div>
    <div id="plot6" class="plot"></div>
  </div>

  <div class="plot-card">
    <div class="plot-title">Community Care</div>
    <div id="plot7" class="plot"></div>
  </div>

  <div class="plot-card">
    <div class="plot-title">Elective Capacity</div>
    <div id="plot8" class="plot"></div>
  </div>

  <div class="plot-card">
    <div class="plot-title">Population Pressure</div>
    <div id="plot9" class="plot"></div>
  </div>

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</div>

<div class="plot-card">
<div class="plot-title">Energy Costs</div>
<div id="plot10" class="plot"></div>
</div>

<div class="plot-card">
<div class="plot-title">Inflation</div>
<div id="plot11" class="plot"></div>
</div>

<div class="plot-card">
<div class="plot-title">Policy Effectiveness</div>
<div id="plot12" class="plot"></div>
</div>

</div>

<!-- DOCUMENTATION -->

<details class="docs">
<summary>Model Documentation</summary>
<div class="docs-content">

This dashboard architecture is optimized for large-scale system dynamics
models with:

<ul>
<li>High parameter counts</li>
<li>Large causal structures</li>
<li>Many simultaneous scenarios</li>
<li>Policy experimentation</li>
<li>Sensitivity analysis</li>
<li>Dense comparative visualization</li>
</ul>

The interface prioritizes:

<ul>
<li>Minimal scrolling</li>
<li>High information density</li>
<li>Fast interaction</li>
<li>Compact visualization</li>
<li>Research-oriented workflows</li>
</ul>

</div>
</details>

</div>

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</div>

<script>

/* SLIDER VALUE UPDATES */

for(let i=1;i<=13;i++){

    const s=document.getElementById('s'+i);
    const v=document.getElementById('s'+i+'v');

    s.addEventListener('input',()=>{
        v.innerHTML=s.value;
        updatePlots();
    });

}

/* SIMULATION */

let currentDay=0;
let timer=null;

function generateSeries(multiplier=1){

    let x=[];
    let y=[];

    for(let t=0;t<400;t++){

        x.push(t);

        y.push(
            100+
            50*Math.sin(t/40)+
            t*0.2*multiplier+
            Math.random()*5
        );

    }

    return {x,y};

}

/* COMMON PLOT LAYOUT */

function compactLayout(){

    return {

```

```

margin:{
  t:10,
  l:30,
  r:5,
  b:20
},

paper_bgcolor: 'white',
plot_bgcolor: 'white',

xaxis:{
  showgrid: false,
  tickfont: {size:8},
  zeroline: false
},

yaxis:{
  showgrid: true,
  gridcolor: '#f0f0f0',
  tickfont: {size:8},
  zeroline: false
},

showlegend: false

};

}

/* PLOT RENDERING */

function renderPlot(id,mult){

  const s=generateSeries(mult);

  Plotly.react(
    id,
    [{
      x:s.x,
      y:s.y,
      mode: 'lines',
      line:{width:1.5}
    }],
    compactLayout(),
    {
      responsive: true,
      displayModeBar: false
    }
  );
};

```

```

}

function updatePlots(){

    renderPlot('plot1',1);
    renderPlot('plot2',1.1);
    renderPlot('plot3',0.7);
    renderPlot('plot4',0.9);
    renderPlot('plot5',1.2);
    renderPlot('plot6',1.3);
    renderPlot('plot7',0.8);
    renderPlot('plot8',1.0);
    renderPlot('plot9',1.4);
    renderPlot('plot10',1.1);
    renderPlot('plot11',0.6);
    renderPlot('plot12',0.5);

}

/* CONTROLS */

function runSimulation(){

    clearInterval(timer);

    timer=setInterval(=>{

        currentDay+=5;

        document.getElementById('simDay').innerHTML=currentDay;

        updatePlots();

    },300);

}

function pauseSimulation(){
    clearInterval(timer);
}

function resetSimulation(){

    currentDay=0;

    document.getElementById('simDay').innerHTML=0;

    updatePlots();

}

```

```
function randomizeParameters(){

    for(let i=1;i<=13;i++){

        const s=document.getElementById('s'+i);

        s.value=Math.random()*s.max;

        document.getElementById('s'+i+'v').innerHTML=
            Number(s.value).toFixed(2);

    }

    updatePlots();

}

function exportData(){
    alert('Export system can be connected here.');
```

```
}

/* INITIALIZE */

updatePlots();

</script>

</body>
</html>
```